

Flutterwave Risk Analytics Assessment

The Scenario

You have been provided with a sample dataset (flutterwave_synthetic_txs.csv) containing recent transaction records. Your objective is to process this data, engineer features, and build a baseline model to identify fraudulent transactions.

Data Dictionary

- transaction_id: A unique, primary key identifier for each transaction.
- user_id: The identifier for the user initiating the transaction.
- timestamp: The time the transaction was processed.
- amount_usd: The transaction value in USD.
- card_country: The country of origin for the card used.
- merchant_category: The category of the merchant receiving the funds.
- device_id: The identifier of the device used for the transaction.
- dispute_status: The current resolution state of the transaction with the issuing bank.
- is_fraud: The target variable indicating that the transaction resulted in a fraud chargeback (1 = Fraudulent, 0 = Legitimate).

Part 1: Data Engineering & SQL

Before modeling, we need to extract features. Write a single, production-ready SQL query (assume a PostgreSQL or Redshift environment) that could run against this raw data table to calculate the following for *every user* at the exact moment of each transaction:

1. The total amount_usd transacted by that user in the preceding 24 hours.
2. The distinct count of card_country used by that user in the preceding 1 hour.

(Note: You do not need a database to run this; just provide the written SQL query in your notebook and explain your performance considerations for running this at scale).

Part 2: Fraud Modeling

Using Python, load the provided CSV dataset and build a predictive model to identify is_fraud.

- Perform necessary data cleaning and exploratory data analysis.
- Engineer any features you believe are relevant.
- Train your choice of machine learning algorithm.
- Evaluate and document your model's performance metrics.

Part 3: Business Strategy

Our Product team has raised concerns that the new fraud model you just built might be too aggressive, potentially blocking legitimate high-value transactions and creating friction for our users. Conversely, the Compliance & Risk team insists that letting too much fraud slip through will result in severe network fines.

Based on the specific outputs of the model you built in Part 2, propose a concrete, data-driven strategy to balance these two conflicting business needs without simply "retraining the model." Include this in your summary document.